

COLLEGE EVENT FEEDBACK ANALYSIS

Data Science & Analytics Internship Project

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Objective

This project focuses on analyzing college event feedback using advanced data science and analytics techniques. By processing student responses, we aim to uncover deep insights regarding satisfaction levels, sentiment patterns, and specific areas for operational improvement. The analysis bridges the gap between raw data and actionable strategies.



Analyze student feedback collected systematically after various college events.



Understand overall satisfaction levels using numerical ratings and statistical aggregation.



Apply basic Natural Language Processing (NLP) techniques to interpret textual sentiment.



Identify specific strengths and weaknesses of events based on qualitative feedback data.



Visualize insights through charts and graphs to help organizers improve future events.



Demonstrate the practical application of data science concepts in a real-world educational scenario.

Dataset Description

The dataset utilized in this project represents feedback collected systematically from students after attending various college events. The data was gathered via a structured Google Form mechanism, capturing both quantitative satisfaction ratings (numerical) and descriptive metadata regarding the specific courses and feedback configurations.



DATASET NAME

FUTURE_DS_03_Dataset



FORMAT

Structured CSV



SOURCE

Simulated / Google Form



PURPOSE

Feedback Analysis

COLUMN NAME	DESCRIPTION	DATA TYPE
SN	Serial number for each feedback record	Integer
Total Feedback Given	Total number of feedback responses collected	Integer
Total Configured	Total expected feedback entries	Integer
Questions	Number of questions in the feedback form	Integer
Weightage 1	Count of responses with rating 1	Integer
Weightage 2	Count of responses with rating 2	Integer
Weightage 3	Count of responses with rating 3	Integer
Weightage 4	Count of responses with rating 4	Integer
Weightage 5	Count of responses with rating 5	Integer
Average / Percentage	Average score or percentage satisfaction	Float
Course Name	Name of the course or event	Text
Basic Course	Category or type of course/event	Text

Tools & Technologies Used

To ensure efficiency and accuracy in this project, a comprehensive stack of modern data science tools was utilized. From cloud-based environments to specialized libraries for visualization and NLP, each tool played a critical role in the analytical workflow.



Python

The core programming language used for the entire project. Its rich ecosystem of libraries enabled seamless data processing, statistical analysis, and algorithm implementation.



Google Colab

A cloud-based IDE used to execute Python code without local setup, offering free GPU access and easy sharing.



Jupyter Notebook

Used for organizing the workflow, combining executable code, visualizations, and rich text explanations in one document.



Pandas

The primary library used for data manipulation, cleaning, and structuring raw CSV data into DataFrames.



Matplotlib

Used for creating foundational charts and graphs to visualize distributions and basic trends in the feedback data.



Seaborn

An advanced visualization library used to create statistical graphics like heatmaps and aesthetically pleasing bar plots.



TextBlob / VADER

Natural Language Processing libraries utilized to perform sentiment analysis on the textual feedback comments.

Methodology



Data Cleaning & Preparation

Removing errors, handling missing values.



Exploratory Data Analysis (EDA)

Understanding patterns and distributions.



Numeric Analysis

Analyzing ratings and feedback scores.



Sentiment Analysis

Identifying positive, neutral, negative opinions.



Correlation Analysis

Finding relationships between key variables.



Visualizations

Presenting insights using charts and graphs.

Data Cleaning & Preparation

Data cleaning is a foundational step in ensuring analysis accuracy. The raw feedback dataset was rigorously processed to remove inconsistencies, standardize formats, and handle missing entries, transforming it into a reliable structure for statistical modeling.

✓ Checked dataset structure

✓ Renamed columns for consistency

✓ Handled missing values (NaN)

✓ Corrected data types

✓ Removed duplicate records

✓ Created derived features

cleaning_script.py

```
# 1. Load Dataset
df = pd.read_csv("FUTURE_DS_03_Dataset.csv")

# 2. Standardize Column Names
df.columns = df.columns.str.strip().str.lower().str.replace(" ", "_")

# 3. Handle Missing Values (Fill 0 for numeric)
numeric_cols = df.select_dtypes(include=["int64", "float64"]).columns
df[numeric_cols] = df[numeric_cols].fillna(0)

# 4. Remove Duplicates
df = df.drop_duplicates()

# 5. Feature Engineering: Total Ratings
df["total_ratings"] = (
    df["weightage_1"] + df["weightage_2"] +
    df["weightage_3"] + df["weightage_4"] +
    df["weightage_5"]
)
```

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to examine the structure, statistical characteristics, and underlying patterns of the college event feedback dataset. This step helps identify satisfaction trends, rating behavior, and course-level differences before applying advanced analytical techniques.



Dataset Structure

Understanding rows, columns, and types



Summary Statistics

Mean, median, and overall spread



Satisfaction Dist.

Identifying score concentration



Top Courses

Courses with highest feedback ratings



Rating Behavior

Student rating preference analysis



Category Dist.

Course category participation overview



Histogram



Top Courses



Ratings



Categories

eda_script.py

```
# 1. Statistical Summary
df.describe()

# 2. Histogram of Average Scores
sns.histplot(avg_clean, bins=10, kde=True)

# 3. Top 10 Performing Courses
top_courses = df.groupby("course_name")["total_ratings"].sum().head(10)

# 4. Rating Bar Chart
weightage_sum.plot(kind="bar")

# 5. Category Pie Chart
df["basic_course"].value_counts().plot(kind="pie")
```

Numeric Analysis

Numeric analysis was conducted to quantitatively evaluate student feedback using rating-based metrics. By computing weighted average scores and analyzing rating distributions, this step provides objective insights into course performance and student satisfaction levels.



Weighted Calculation

Deriving meaningful scores from 1–5 ratings.



Summary Statistics

Mean, range, and distribution of ratings.



Top Performers

Courses with highest weighted ratings.



Rating Variability

Consistency and spread of feedback scores.

KEY NUMERIC INSIGHTS

- Weighted ratings provide a more reliable measure than raw averages.
- Higher concentration of ratings at levels 4 and 5 indicates positive feedback.
- Some courses show greater variability in satisfaction levels.

numeric_analysis.py

```
# 1. Compute Weighted Average Rating
df["weighted_average_rating"] = df["weighted_score_sum"] / df["rating_count"]

# 2. Statistical Summary of Ratings
df["weighted_average_rating"].describe()

# 3. Top 10 Courses by Mean Rating
df.groupby("course_name")["weighted_average_rating"].mean().head(10)

# 4. Visualization: Rating Totals Bar Chart
rating_totals.plot(kind="bar")

# 5. Visualization: Boxplot of Valid Scores
sns.boxplot(x=valid_scores)
```

Sentiment Analysis

Sentiment analysis was performed to classify overall student feedback into positive, neutral, and negative categories. Since textual comments were not available, sentiment was derived logically from rating behavior using weighted average scores to approximate satisfaction levels.



Positive Sentiment

High satisfaction levels.

Rating ≥ 4



Neutral Sentiment

Moderate satisfaction.

2.5 ≤ Rating < 4



Negative Sentiment

Low satisfaction levels.

Rating < 2.5

KEY SENTIMENT INSIGHTS

- ✓ Majority of courses fall under positive sentiment, indicating healthy engagement.
- ✓ Neutral sentiment suggests specific operational areas require minor improvements.
- ✓ Negative sentiment is minimal but highlights critical pain points.

SENTIMENT DIST.



Positive | Neutral | Negative

```
sentiment_logic.py

# 1. Apply Sentiment Classification Logic
df["sentiment"] = df["weighted_average_rating"].apply(assign_sentiment)

# 2. Check Overall Sentiment Counts
df["sentiment"].value_counts()

# 3. Analyze Sentiment by Course Category
df.groupby(["basic_course", "sentiment"]).size().unstack()
```

Correlation Analysis

Correlation analysis was conducted to examine the relationships between different numerical variables in the dataset. This analysis helps understand how rating components influence overall satisfaction and whether strong linear relationships exist among feedback metrics.

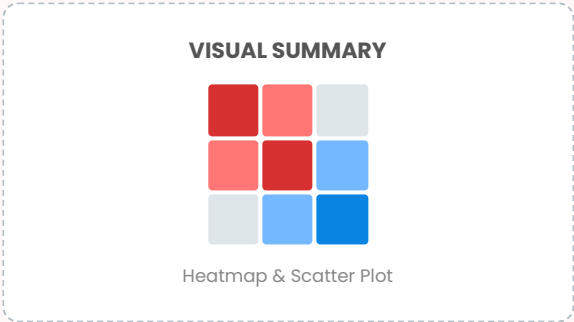
**Rating Relations**
Interdependence between 1–5 ratings.

**Total vs Satisfaction**
Volume vs quality relationship.

**Correlation Strength**
Positive, negative, & weak associations.

**Insight Discovery**
Identifying influential variables.

- KEY OBSERVATIONS**
- Higher weightage ratings (4 & 5) show strong positive correlation with satisfaction.
 - Total ratings moderately influence overall average rating.
 - Lower ratings show weaker or negative correlations.



```
correlation_matrix.py

# 1. Compute Correlation Matrix
correlation_matrix = corr_df.corr()

# 2. Visualize Heatmap
sns.heatmap(correlation_matrix, annot=True, cmap="coolwarm")

# 3. Visualize Scatter Plot (Volume vs Rating)
sns.scatterplot(
    x="total_ratings",
    y="weighted_average_rating",
    data=df
)
```

Visualizations (1/3)



Overall Distribution of Ratings (1–5)

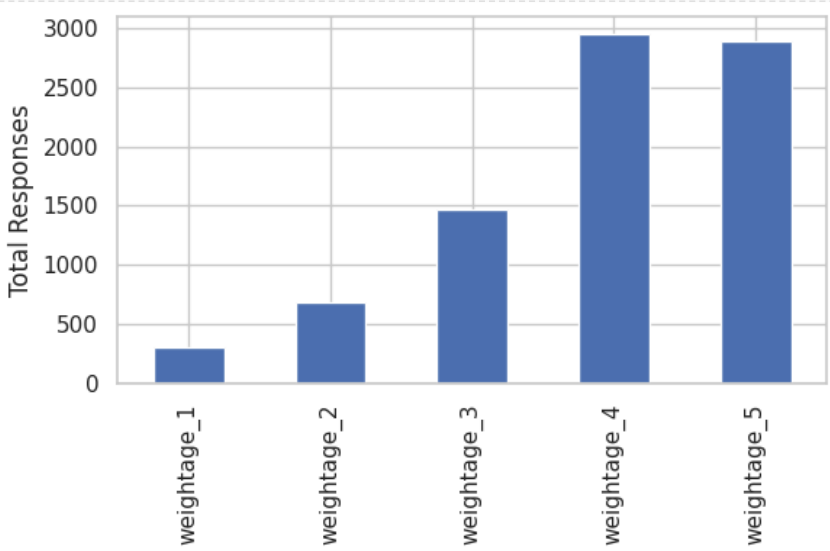


Figure 1: Overall student rating distribution pattern



Top 10 Courses (Avg Rating)

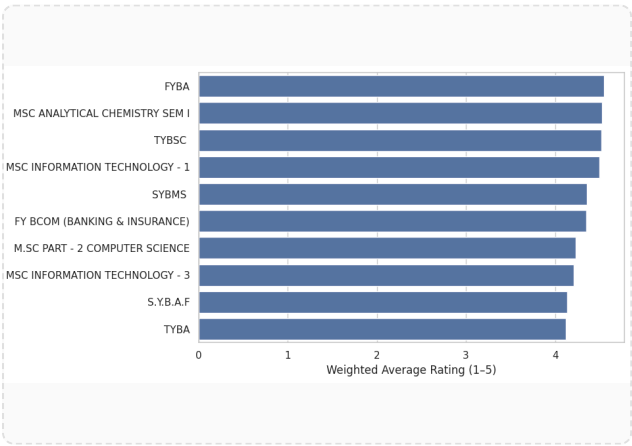


Figure 2: Highest rated courses by satisfaction



Top 10 Courses (Volume)

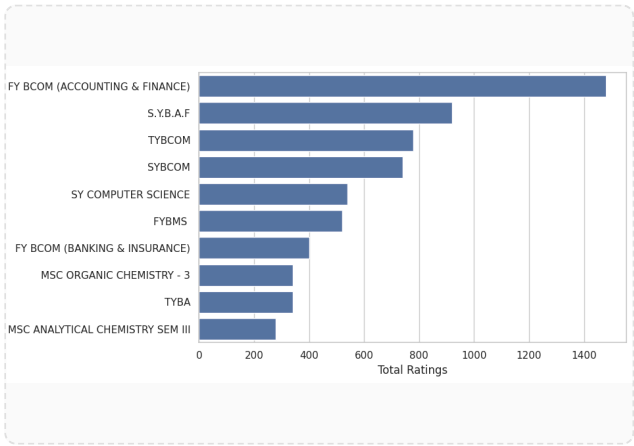


Figure 3: Courses receiving maximum feedback

Visualizations (2/3)



Overall Sentiment

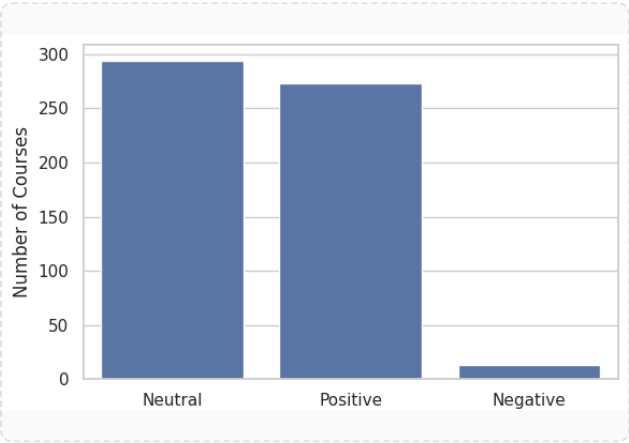


Figure 4: Overall sentiment classification



Correlation Heatmap

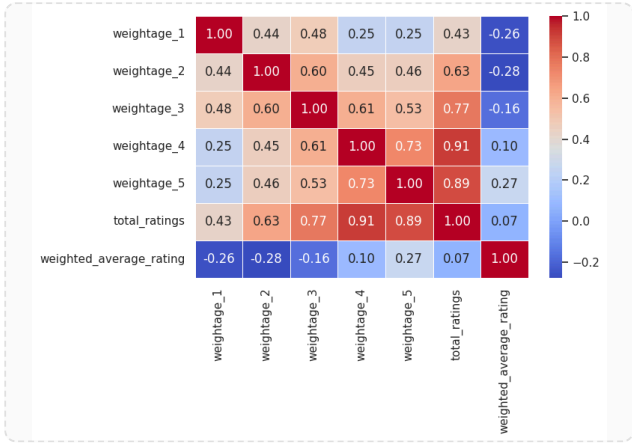


Figure 6: Correlation between feedback variables



Sentiment by Course Category

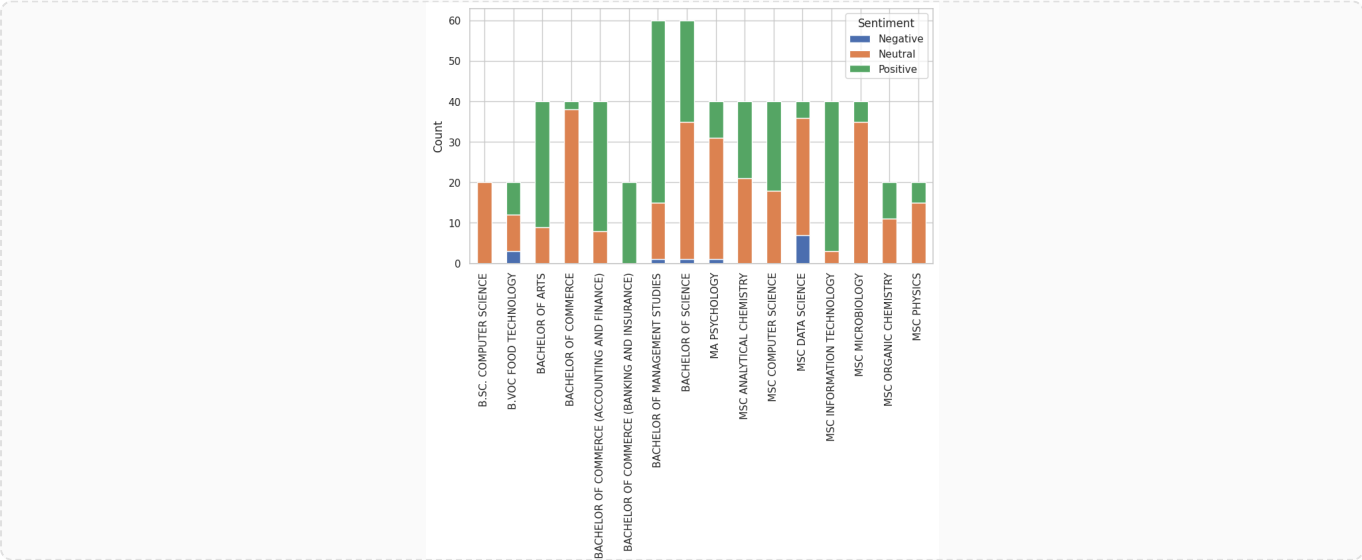


Figure 5: Sentiment comparison across course categories

Visualizations (3/3)



Total Ratings vs Avg Satisfaction

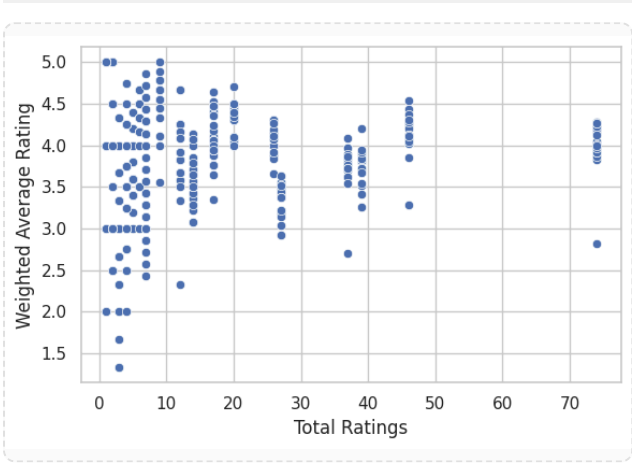


Figure 7: Relationship between feedback volume and satisfaction



Distribution of Course Categories

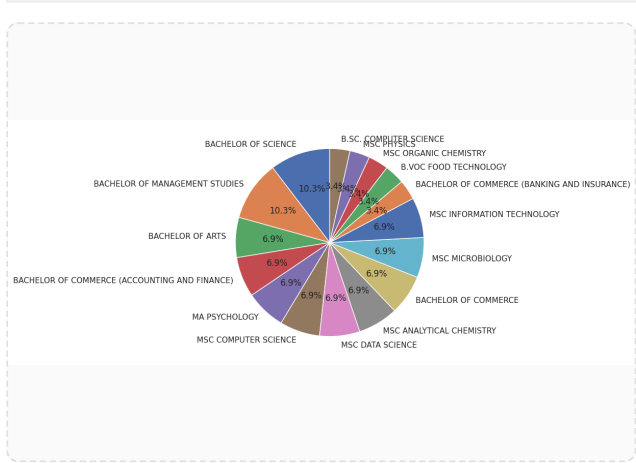


Figure 8: Proportion of different course categories



Variation in Weighted Average Ratings

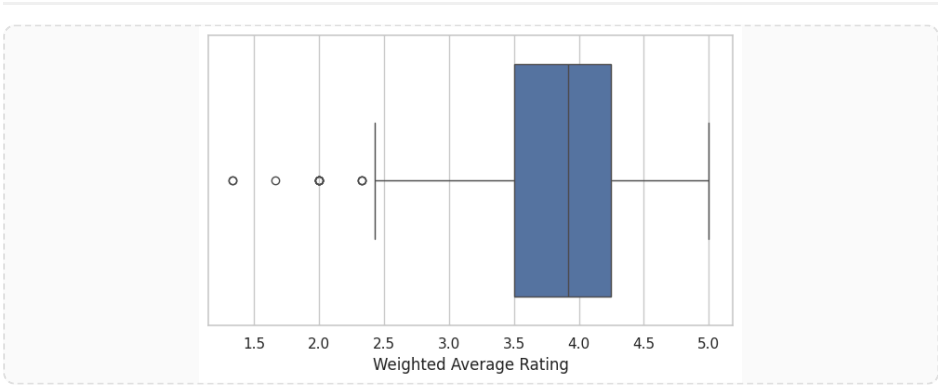


Figure 9: Spread of weighted satisfaction scores

Key Insights

This section summarizes the most important insights derived from data cleaning, exploratory analysis, numeric evaluation, sentiment analysis, correlation analysis, and visual exploration of the college event feedback dataset. These insights reflect overall student satisfaction patterns and key performance indicators for college events.



High Overall Student Satisfaction

Majority of feedback reflects positive to neutral sentiment, indicating successful event planning and execution.



Strong Preference for Higher Ratings

Ratings of 4 and 5 dominate the feedback distribution, showing highly favorable student experiences.



Top-Performing Courses Identified

Certain courses consistently achieve higher weighted average ratings, indicating strong student engagement.



Feedback Volume Varies by Course

Some courses receive significantly more feedback, suggesting higher participation rates or popularity.



Sentiment Varies Across Categories

Course categories differ in sentiment distribution, highlighting areas of excellence and potential improvement.



Key Metrics Are Interrelated

Strong correlations exist between rating weightages and total feedback, influencing overall satisfaction scores.

Recommendations & Conclusion

Based on the insights obtained from the analysis of college event feedback, this section presents actionable recommendations to improve future events, followed by a concise conclusion summarizing the overall project outcomes and learning experience.



Enhance High-Impact Formats

Focus on event formats and courses that consistently receive high satisfaction ratings to maximize engagement.



Improve Neutral Feedback

Analyze neutral sentiment responses deeply to identify specific opportunities for incremental improvements.



Encourage Participation

Increase student engagement in feedback collection processes to ensure representative and diverse insights.



Data-Driven Planning

Leverage specific feedback metrics to guide event scheduling, content curation, and resource allocation.



Regular Feedback Monitoring

Continuously track satisfaction trends over subsequent semesters to measure the impact of changes and improvement over time.

CONCLUSION

This project successfully demonstrated how data science and analytics techniques can be applied to real-world student feedback to extract meaningful insights. Through systematic data cleaning, analysis, visualization, and interpretation, the project highlights the value of data-driven decision-making in enhancing college event experiences.

THANK YOU

Thank you for reviewing this internship project and taking the time to explore the insights derived from data-driven analysis.

— CREATED BY —

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